Skills Summary

Telling Exponential Change from Linear Change

Use this reference while completing your worksheet or when studying.

Skill 1 — The differences-and-ratios test

Test a table over equal input steps:

Linear — successive outputs have a constant **difference** d; rule $y = a + dx$.

Exponential — successive outputs have a constant **ratio** r; rule $y = a \cdot r^x$.

Do both tests, not one. A table can look "steadily growing" either way; only the constant one tells you which.

Example: Test the table $2,\ 6,\ 18,\ 54$ at inputs $0, 1, 2, 3$.
Step 1. The inputs step by one each time, so both tests are fair — run the ratio test and the difference test and see which comes out constant.
Step 2. Ratios: $6 \div 2 = 3$, and $18 \div 6 = 3$ as well, so the ratio holds.
Step 3. Differences: $6 - 2 = 4$, then $18 - 6 = 12$ — not constant.
ratio $=$ **3**, differences **4** then **12**: exponential, $y = 2 \cdot 3^x$.

Try it: Test $7,\ 11,\ 15,\ 19$. Find the first difference and the first ratio.

check: difference **4** (constant, so linear), ratio $\frac{11}{7}$

Skill 2 — Evaluating each kind of rule

Linear: $a + dx$ — multiply the step by the input, then add the start.

Exponential: $a \cdot r^x$ — raise the multiplier to the input *first*, then multiply by the start. The exponent applies to r alone, never to a.

Percentages: a rise of $p\%$ makes $r = 1 + \frac{p}{100}$; a fall of $p\%$ makes $r = 1 - \frac{p}{100}$. So 15% growth is $r = 1.15$ and 12% decay is $r = 0.88$.

Example: $L(x) = 12 + 5x$ and $E(x) = 12 \cdot 1.5^x$. Find both at $x = 4$.
Step 1. One rule adds a fixed amount each step and the other multiplies by a fixed factor, so the arithmetic order differs: multiply-then-add against power-then-multiply.
Step 2. $L(4) = 12 + 5(4) = 12 + 20$.
Step 3. $E(4)$: first $1.5^4 = 5.0625$, then 12×5.0625.
$L(4) = \mathbf{32}$ and $E(4) = \mathbf{60.75}$

Try it: $L(x) = 20 + 7x$ and $E(x) = 20 \cdot 1.2^x$. Find both at $x = 3$.

check: $L(3) = \mathbf{41}$, $E(3) = \mathbf{34.56}$

Skill 3 — Which model wins in the long run

The headline fact: a growing exponential eventually passes *any* linear rule, however big the linear step and however small the start. Early values prove nothing — the linear rule is often ahead for the first few inputs.

Why: the linear rule adds the same amount every step; the exponential rule adds a percentage of a total that keeps getting larger, so its own step grows.

Example: $f(x) = 30 + 10x$ and $g(x) = 3 \cdot 2^x$. Compare them at $x = 4$ and at $x = 6$.
Step 1. One comparison cannot show a crossover, so test an early input and a later one and watch the order swap.
Step 2. At $x = 4$: $30 + 40$ against $3 \cdot 16$ — the linear rule leads.
Step 3. At $x = 6$: $30 + 60$ against $3 \cdot 64$ — the exponential rule has taken over.

$f(4) = \mathbf{70}$, $g(4) = \mathbf{48}$; $f(6) = \mathbf{90}$, $g(6) = \mathbf{192}$

Try it: $f(x) = 50 + 5x$ and $g(x) = 2 \cdot 3^x$. Compare them at $x = 5$.

check: $f(5) = \mathbf{75}$, $g(5) = \mathbf{486}$